

«Rejection Sampling» ≡

Rejection, Reweight.
Repeat

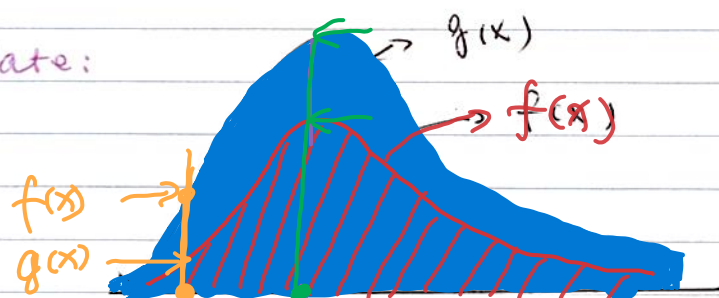
First. Review

$$I = \int a(\theta) f(\theta) d\theta \quad \theta \sim f(\theta)$$

$$\hat{I} = \frac{1}{N} \sum_{i=1}^N a(\theta_i) \quad / N \quad (\text{Monte Carlo method})$$

Overall acceptance Rate:

$$= \frac{\int_{-\infty}^{\infty} f(x) dx}{\int_{-\infty}^{\infty} g(x) dx}$$



Rejection Sampling 1:

$g(x) \geq f(x)$ for all x envelop function

for (i in 1 : n)

{ repeat

{ Draw $x \sim \left(g / \int g(x) dx \right)$.

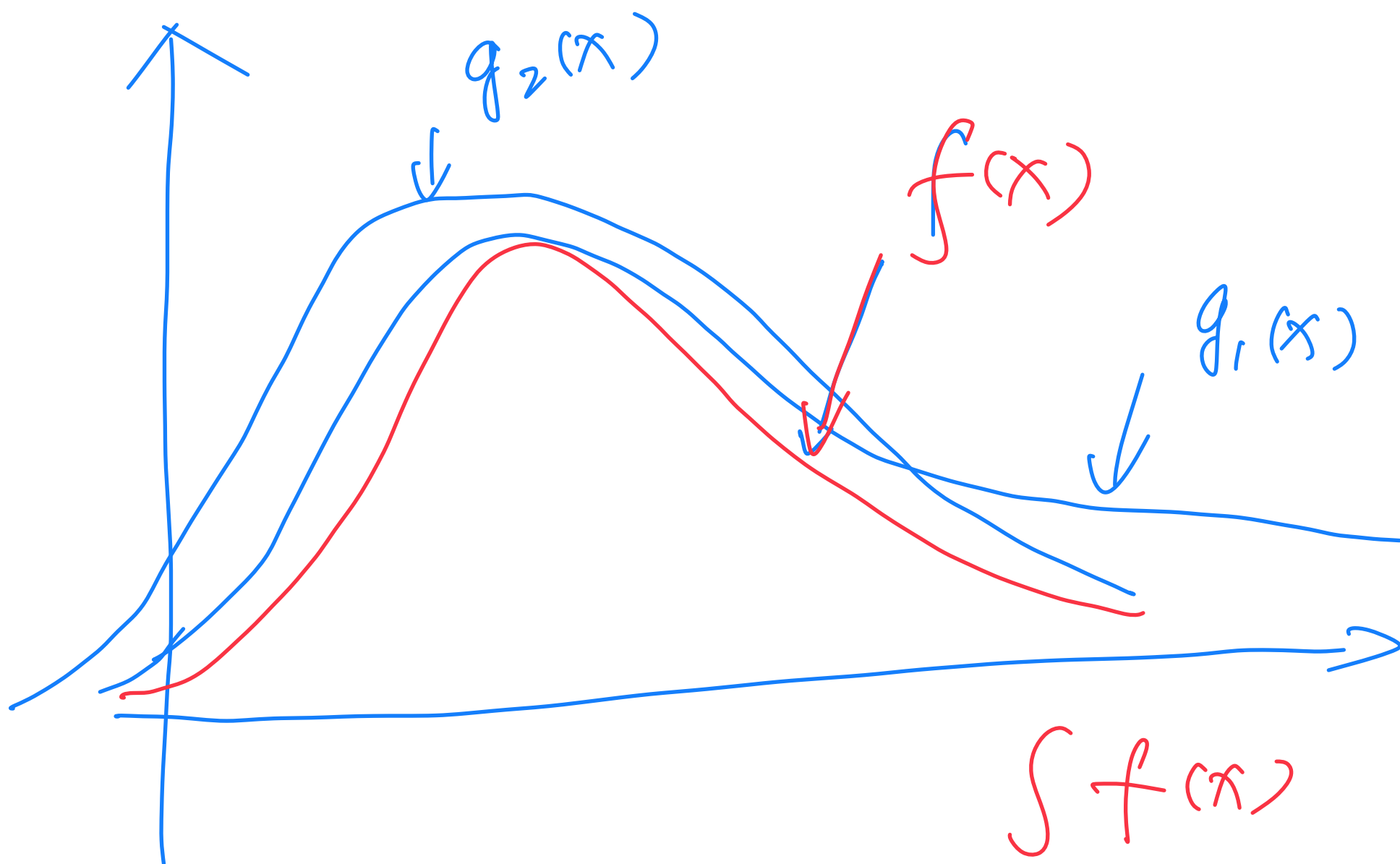
$$r = \frac{f(x)}{g(x)}$$

Draw $v \sim \text{unif}(0, 1)$

if ($v < \frac{f(x)}{g(x)}$) x is accepted

}

}



$$\text{over. Accop.} = \frac{\int f(x)}{\int g(x)}$$

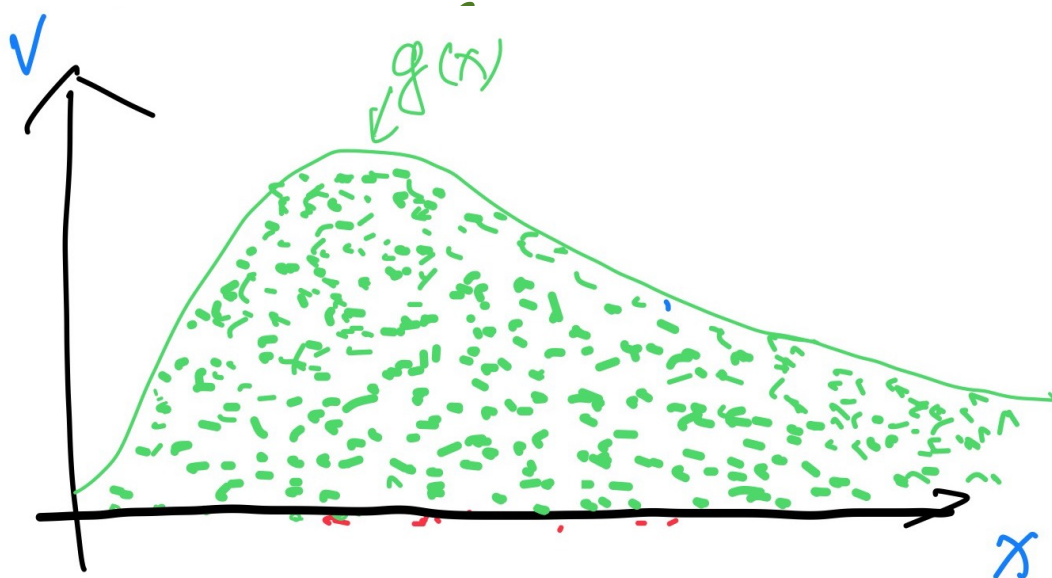
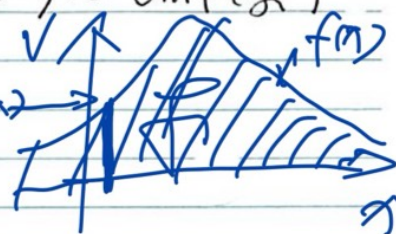
Theorem:

$\Rightarrow$ theorem: Suppose f is a function $f(x) \geq 0$

with $\int f(x) dx < \infty$, if $(x, v) \sim \text{Unif}(G)$

$$G = \{(x, v) \mid 0 < v < f(x)\}$$

then the PDF of x is $\frac{f(x)}{\int f(x) dx}$



$$x \sim g(x), v|x \sim \text{Unif}([0, g(x)])$$

$$\Leftrightarrow (x, v) \sim \text{Unif}(G)$$

$$G = \{(x, v) \mid 0 < v < g(x)\}$$

$$\text{If } (x, v) \in G = \{(x, v) \mid 0 < v < f(x)\}$$

proof: the joint PDF of (X, V)

$$P(x, v) = \frac{1}{\text{area}(f)} \cdot I(0 \leq v < f(x))$$
$$= \frac{1}{\int_{-\infty}^{\infty} f(x) dx} \cdot I(0 \leq v < f(x))$$

$$P(x) = \int_{-\infty}^{\infty} P(x, v) dv$$

$$= \int_0^{f(x)} P(x, v) dv = \int_0^{f(x)} \frac{1}{\int_{-\infty}^{\infty} f(x) dx} dv$$

$$= \frac{1}{\int_{-\infty}^{\infty} f(x) dx} \cdot \int_0^{f(x)} dv = \frac{f(x)}{\int_{-\infty}^{\infty} f(x) dx}$$

Alternative description

Repeat n times:

1) Draw $x \sim q(x)$

2) Draw $V \sim \text{unif}(0, q(x))$

$$[V = \underline{U} \cdot \underline{q(x)}]$$

$$U \sim \text{unif}(0, 1)]$$

3) If $(V < f(x))$ accept x

$$V = U \cdot q(x) < f(x) \Leftrightarrow U < \frac{f(x)}{q(x)}$$

else, go back to 1)

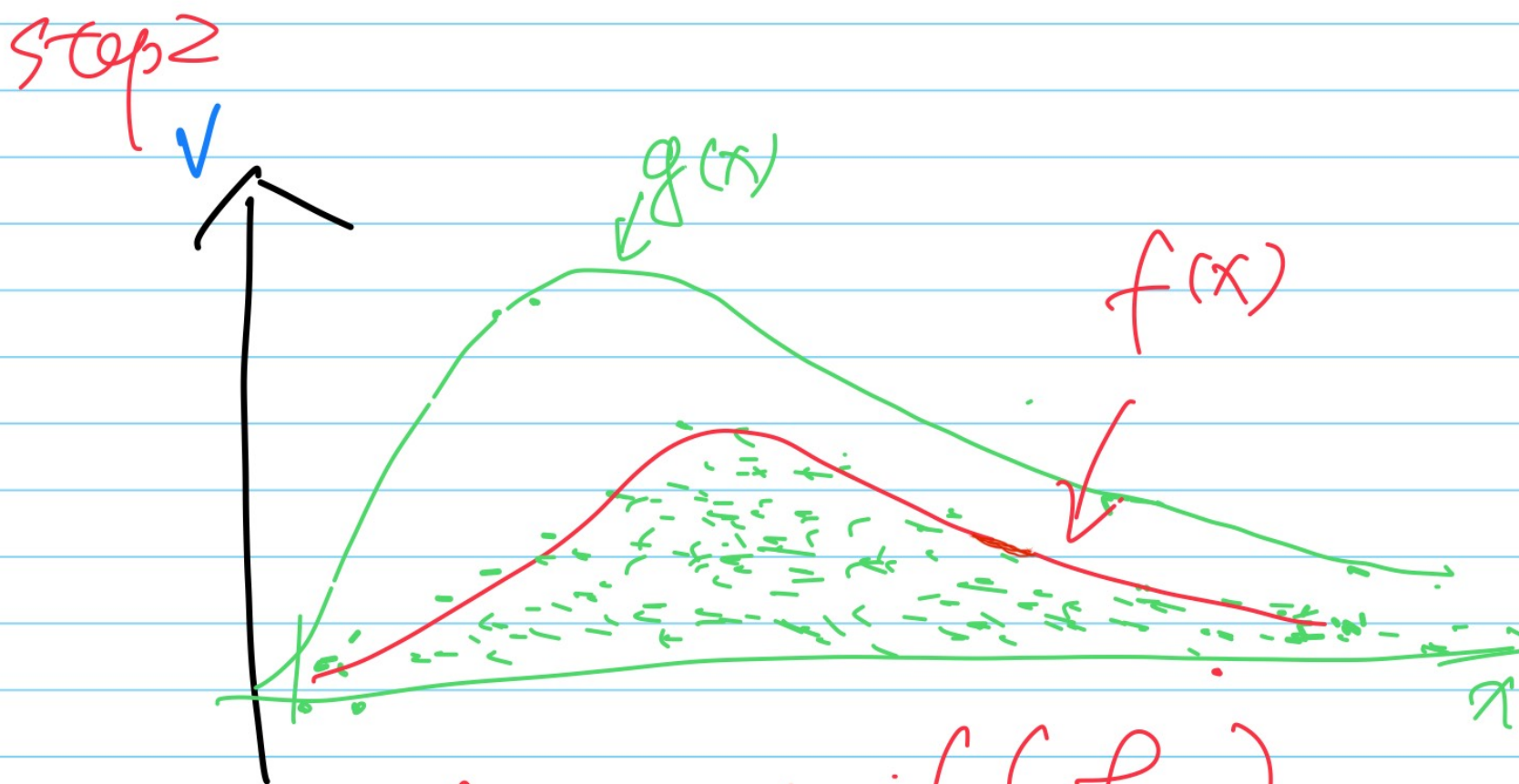
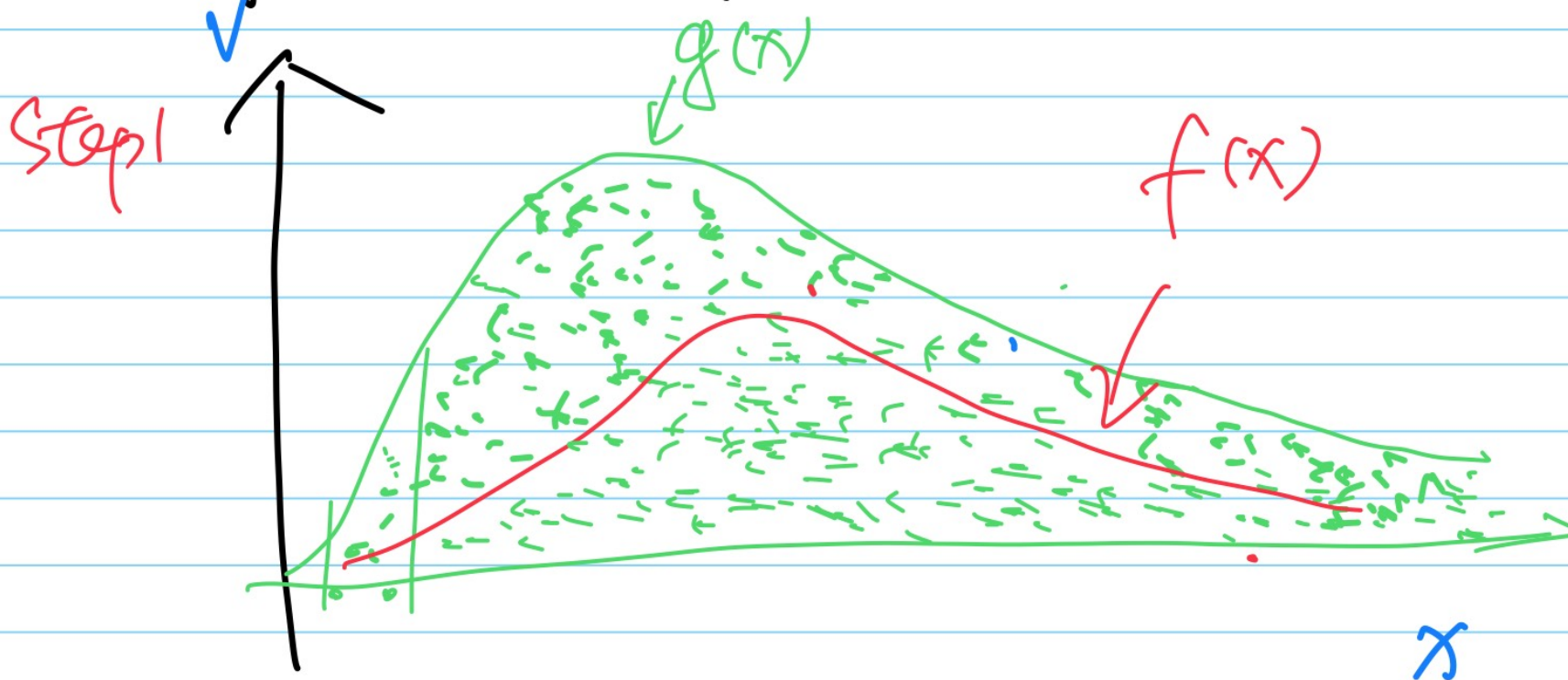
$\Rightarrow$ In 1) and 2) we draw

$$(x, V) \sim \text{Unif}(G)$$

$$\text{where, } G = \{(x, V) \mid V < q(x)\}$$

In 3) choose $(x, V) \in \mathcal{F} = \{(x, V) \mid V < f(x)\}$

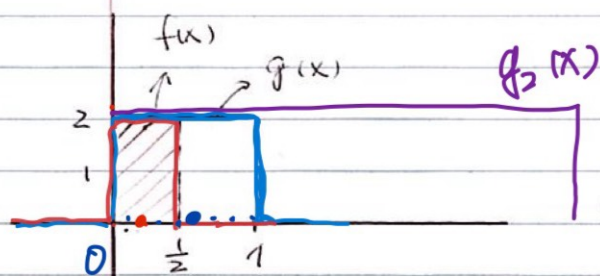
Rejection Sampling (Alternative desc.)



$$(x, v) \sim \text{Unif}(\mathcal{R})$$

$$\mathcal{R} = \{(x, v) \mid 0 < v < f(x)\}$$

example: repeat n times



$$f(x) = 2I(0 \leq x \leq \frac{1}{2})$$

$$g(x) = 2I(0 \leq x \leq 1)$$

step: 1) $X \sim \text{Unif}(0, 1)$

2) $U \sim \text{unif}(0, 1)$

$$3) r = \frac{f(x)}{g(x)}$$

if $(U < r)$ accept X

o/w reject and go back
step 1)

In step(3)

$$r = f/g = \begin{cases} 1 & x < \frac{1}{2} \\ 0 & x > 0 \end{cases}$$

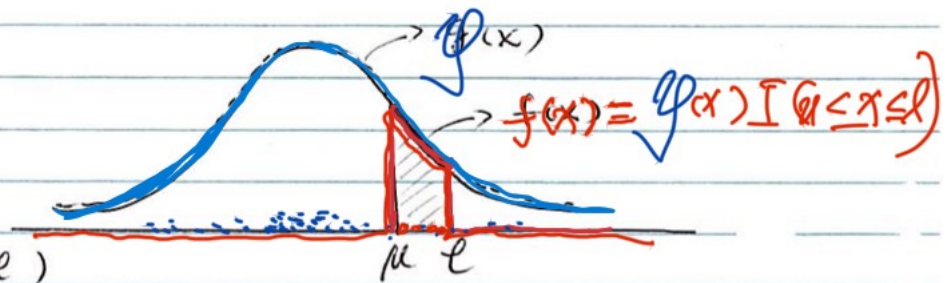
3) can be rewritten 3*)

If $(x < \frac{1}{2})$ accept X

o/w go back to step(1)

$$> \text{overall accept Rate} = \frac{\int_{-\infty}^{\infty} f(x) dx}{\int_{-\infty}^{\infty} g(x) dx} = \frac{1}{2}$$

example 2



$$f(x) = \varphi(x) I(\mu \leq x \leq \ell)$$

$$X \sim f(x) = \varphi(x) I(\mu \leq x \leq \ell)$$

Repeat n times

1) draw $X \sim \varphi(x)$

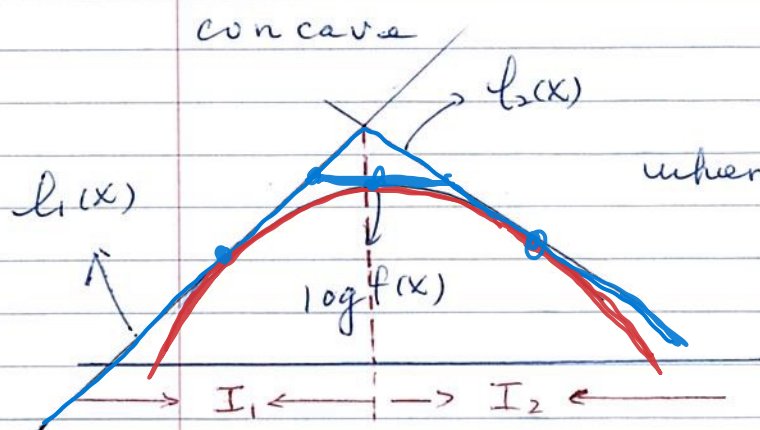
2) if $(X \in (\mu, \ell))$, accept X

o/w go back to 1)

$$\begin{aligned} \text{overall accept rate} &= \frac{\int_{\mu}^{\ell} f(x) dx}{\int_{-\infty}^{\infty} \varphi(x) dx} = \int_{\mu}^{\ell} f(x) dx \\ &= P(\mu \leq x \leq \ell) \end{aligned}$$

Adaptive rejection sampling (ars)

For log-concave $f(x) \Rightarrow \log f(x)$ is



$$g(x) \geq f(x)$$

where,

$$g(x) = \begin{cases} e^{l_1(x)} & x \in I_1 \\ e^{l_2(x)} & x \in I_2 \end{cases}$$

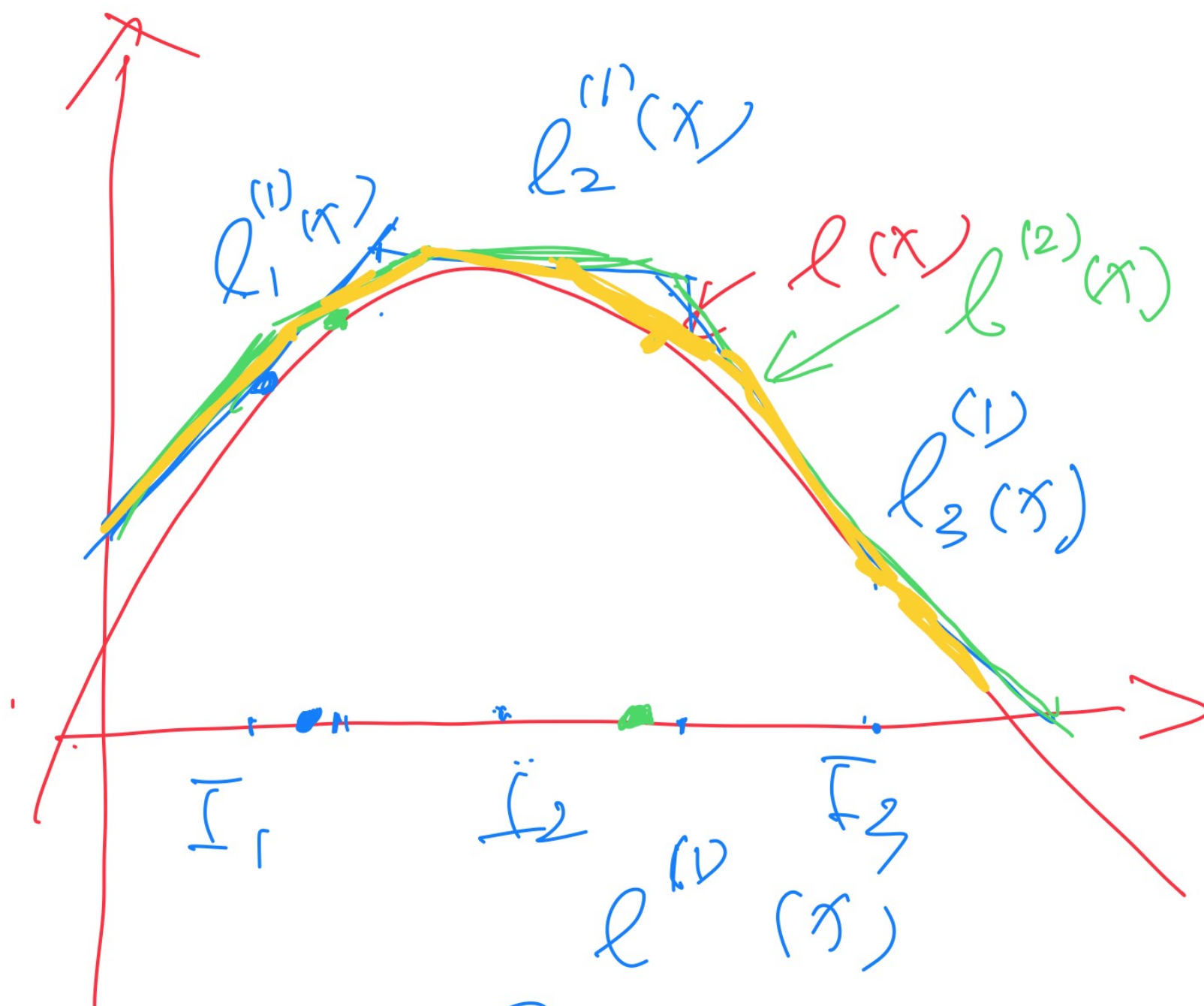
$$\log g(x) = \begin{cases} l_1(x) & x \in I_1 \\ l_2(x) & x \in I_2 \end{cases}$$

$F_g(x) = \int_{-u}^x g(t) dt$ has close form

we can use inverse CDF

$$F_g(x) = \int_{-u}^x g(t) dt = \int_{-u}^x e^{a+bt} dt$$

$$l(x) = \log f(x)$$



$$q_1(x) = e^{l^{(1)}(x)}$$

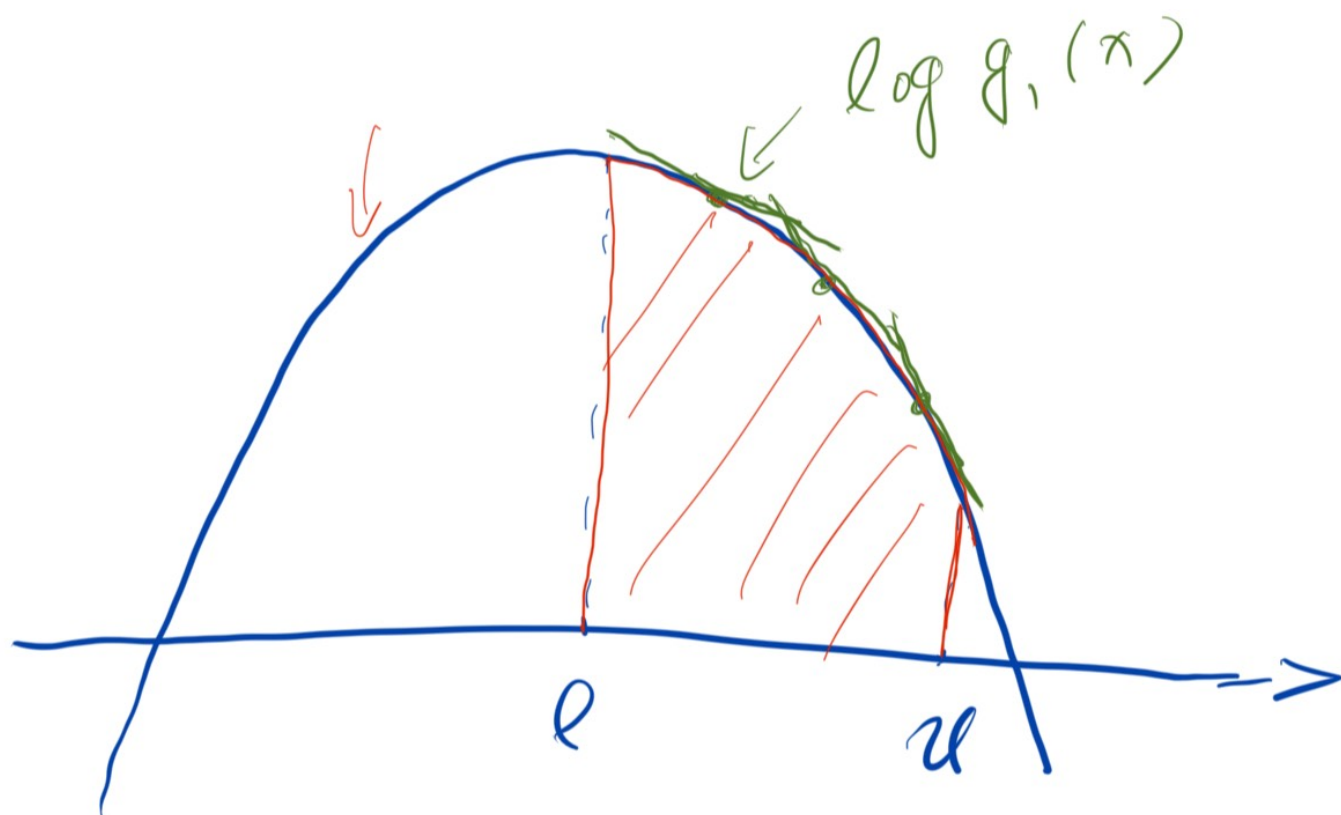
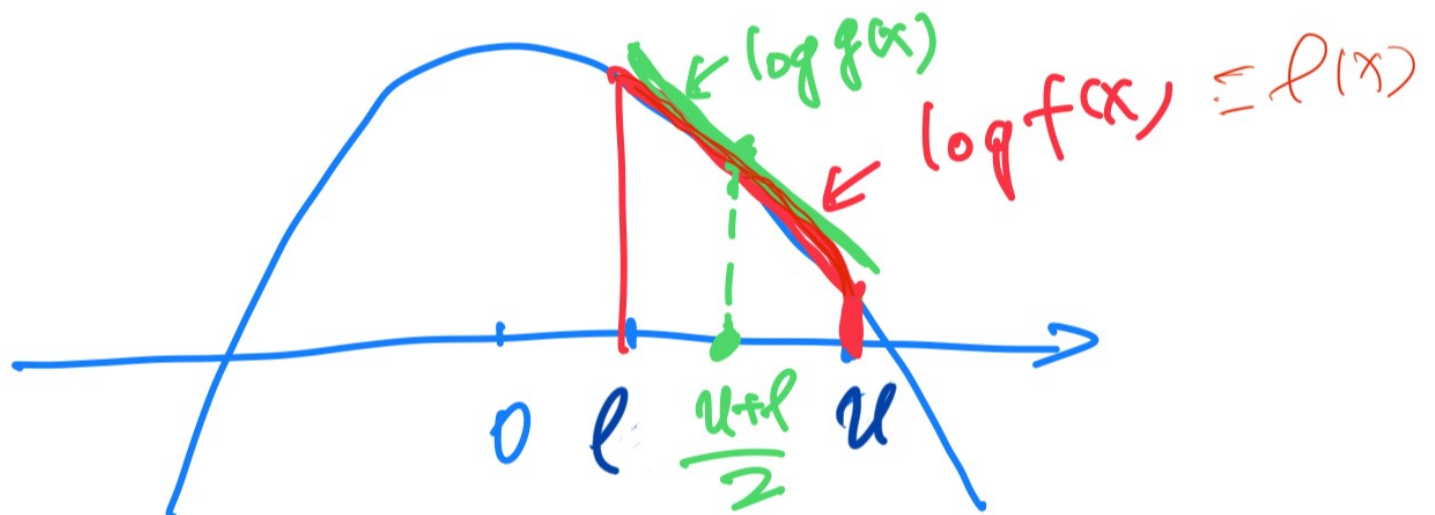
$$q_2(x) = e^{l^{(2)}(x)}$$

Example

Truncated Normal

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \mathbb{I}(u < x < l)$$

$$\log f(x) = -\frac{1}{2}(\log(2\pi)) - \frac{x^2}{2}, \text{ for } l < x < u$$



$$P(l < Z < u)$$